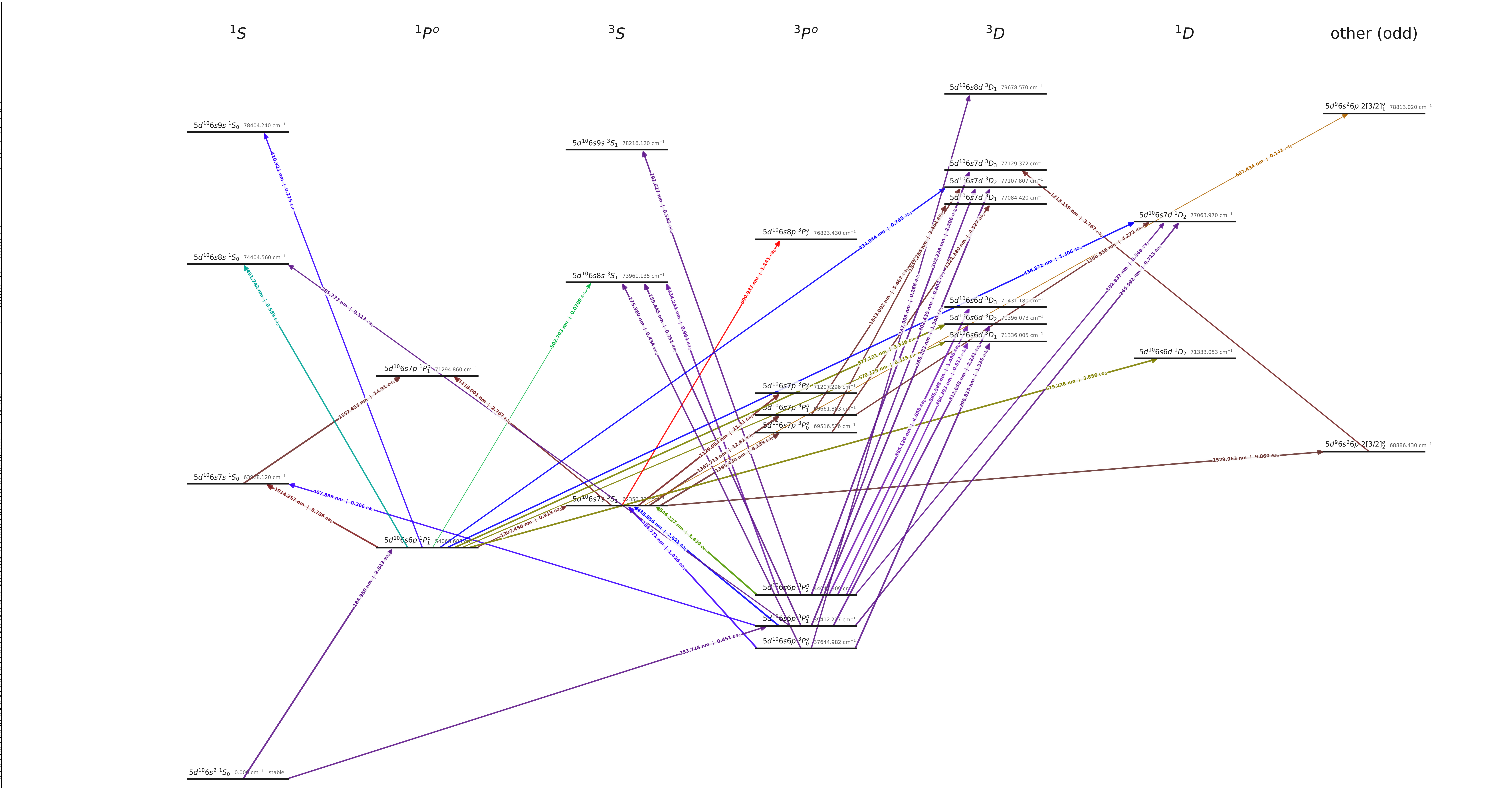


Mercury

Hg (Hg I). Strongest classified transitions below 2 μm (NIST): 27 levels, 45 lines.

Hg ⁺ ionisation limit 84184.15 cm⁻¹ = 10.43750 eV (118.787 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5d ¹⁰ 6s ² ¹ S ₀ – 5d ¹⁰ 6s6p ¹ P ₁ ^o	184.9499	–	1620.9383	2.643	7.460e+08	–	–	–
5d ¹⁰ 6s6p ³ P ₁ ^o – 5d ¹⁰ 6s7d ³ D ₂	265.2832	265.2042	1130.0848	1.340	3.900e+07	–	–	–
5d ¹⁰ 6s6p ³ P ₀ ^o – 5d ¹⁰ 6s6d ³ D ₁	296.8150	296.7283	1010.0315	1.335	4.600e+07	–	–	–
5d ¹⁰ 6s6p ³ P ₂ ^o – 5d ¹⁰ 6s7d ³ D ₃	302.2384	302.1504	991.9072	2.206	5.100e+07	–	–	–
5d ¹⁰ 6s6p ³ P ₁ ^o – 5d ¹⁰ 6s6d ³ D ₂	312.6579	312.5673	958.8513	2.231	6.600e+07	–	–	–
5d ¹⁰ 6s6p ³ P ₂ ^o – 5d ¹⁰ 6s6d ³ D ₃	365.1198	365.0158	821.0797	4.658	1.290e+08	–	–	–
5d ¹⁰ 6s6p ³ P ₀ ^o – 5d ¹⁰ 6s7s ³ S ₁	404.7707	404.6564	740.6476	1.426	2.070e+07	–	–	–
5d ¹⁰ 6s6p ³ P ₁ ^o – 5d ¹⁰ 6s7s ³ S ₁	435.9561	435.8336	687.6666	2.621	5.600e+07	–	–	–
5d ¹⁰ 6s6p ³ P ₂ ^o – 5d ¹⁰ 6s7s ³ S ₁	546.2267	546.0750	548.8425	3.439	4.900e+07	–	–	–
5d ¹⁰ 6s6p ¹ P ₁ ^o – 5d ¹⁰ 6s6d ³ D ₂	577.1210	576.9609	519.4621	3.346	2.360e+07	–	–	–
5d ¹⁰ 6s6p ¹ P ₁ ^o – 5d ¹⁰ 6s6d ¹ D ₂	579.2276	579.0670	517.5728	3.856	3.100e+07	–	–	–
5d ¹⁰ 6s6p ¹ P ₁ ^o – 5d ¹⁰ 6s7s ¹ S ₀	1014.2567	1013.9787	295.5785	3.736	2.710e+07	–	–	–
5d ¹⁰ 6s7s ³ S ₁ – 5d ¹⁰ 6s7p ³ P ₂ ^o	1129.0542	1128.7451	265.5253	11.31	3.600e+07	–	–	–
5d ¹⁰ 6s7s ¹ S ₀ – 5d ¹⁰ 6s7p ¹ P ₁ ^o	1357.4526	1357.0814	220.8493	14.91	6.000e+07	–	–	–
5d ¹⁰ 6s7s ³ S ₁ – 5d ¹⁰ 6s7p ³ P ₁ ^o	1367.7125	1367.3386	219.1926	12.61	4.200e+07	–	–	–
5d ¹⁰ 6s7s ³ S ₁ – 5d ¹⁰ 6s7p ³ P ₀ ^o	1395.4298	1395.0484	214.8388	8.189	5.000e+07	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J||er||J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).

